MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 11

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}, \quad \mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

The variance grows since $\sigma(t)$ is a monotonically increasing function.

Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

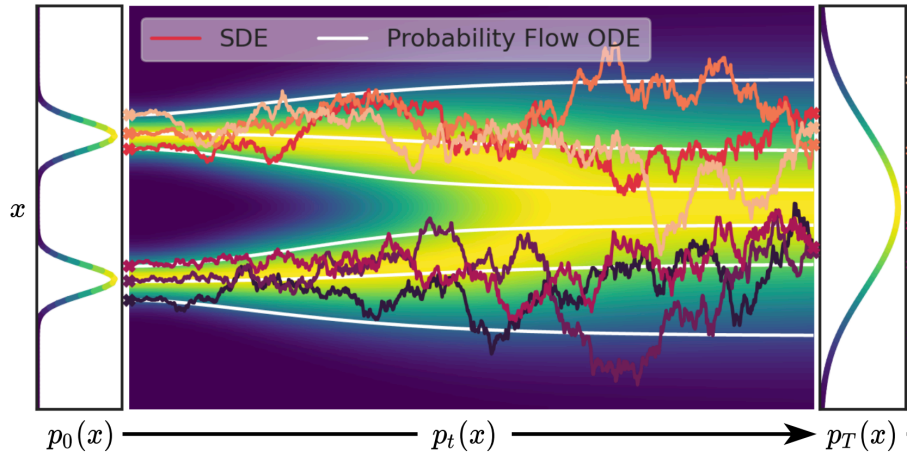
The variance is preserved if $\mathbf{x}(0)$ has unit variance.

Recap of Previous Lecture

Probability Flow ODE

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)dt$$



Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

Reverse ODE

Let $\tau = 1 - t$ ($d\tau = -dt$):

$$d\mathbf{x} = -\mathbf{v}(\mathbf{x}, 1 - \tau)d\tau$$

Reverse SDE

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}(t)$ is a reverse-time Wiener process.

Anderson B. D. O. Reverse-time diffusion equation models, 1982
Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Proof Sketch

- Convert the initial SDE to a probability flow ODE.
- Reverse the probability flow ODE.
- Convert the reversed probability flow ODE back to an SDE.

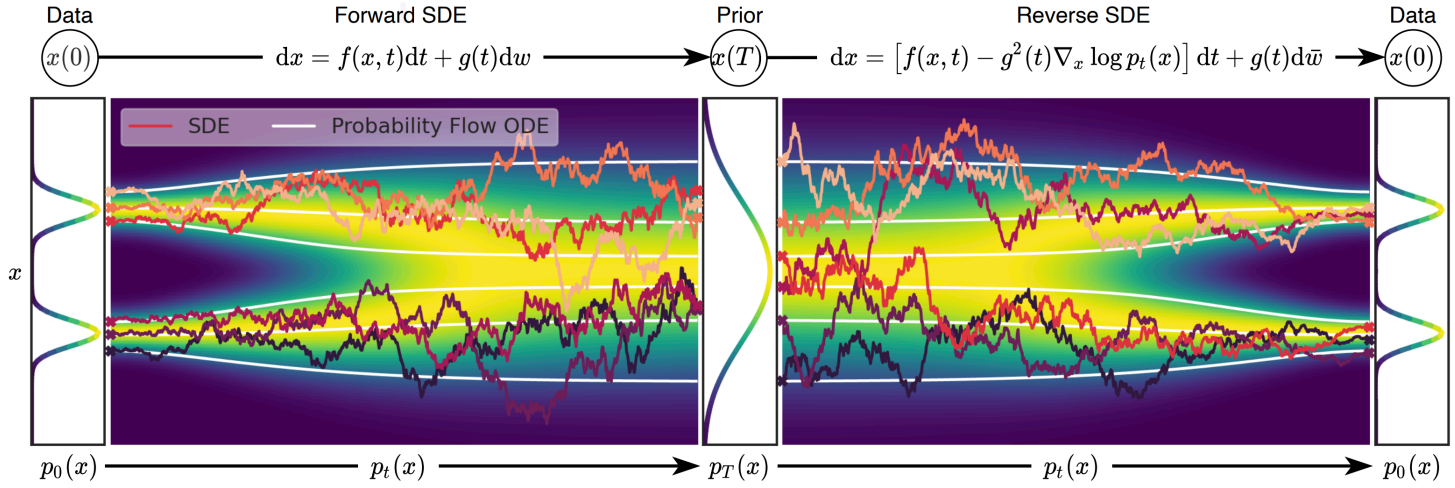
Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t), \quad \text{where} \quad \mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \qquad \textbf{– Probability Flow ODE}$$

$$d\mathbf{x} = \left(\mathbf{v}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)\right)dt + g(t)d\bar{\mathbf{w}} \qquad \textbf{– Reverse SDE}$$



Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t) - \nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$

NCSN

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), [\sigma^2(t) - \sigma^2(0)]\cdot\mathbf{I}\right)$$

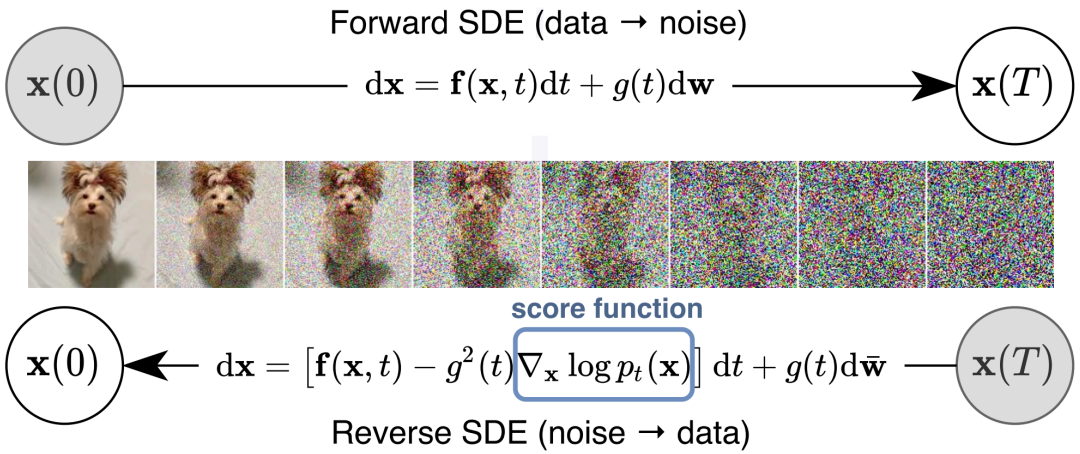
DDPM

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0)e^{-\frac{1}{2}\int_0^t\beta(s)ds}, \left(1 - e^{-\int_0^t\beta(s)ds}\right)\cdot\mathbf{I}\right)$$

Recap of Previous Lecture

Sampling

1. Sample $\mathbf{x}(1) \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the reverse SDE using numerical solvers (`SDESolve`).



- Discretizing the reverse SDE provides ancestral sampling.
- Discretizing the probability flow ODE yields deterministic sampling.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

For additive Gaussian noise $\mathbf{x} = \mathbf{x}_0 + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$:

A Denoiser Parameterizes the Probability Flow ODE

$$\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) = \frac{D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) - \mathbf{x}}{\sigma^2}, \qquad \frac{d\mathbf{x}}{d\sigma} = -\sigma \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, \sigma) = \frac{\mathbf{x} - D_{\boldsymbol{\theta}}(\mathbf{x}, \sigma)}{\sigma}.$$

Decreasing σ moves towards the denoiser prediction.

DDIM Is Euler in Noise Coordinates

Normalize the DDPM state and noise level:

$$\tilde{\mathbf{x}}_t = \frac{\mathbf{x}_t}{\sqrt{\bar{\alpha}_t}}, \qquad \sigma_t = \sqrt{\frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t}}.$$

Deterministic DDIM implements the Euler step using the existing noise predictor.

The ODE field can be obtained from a trained denoiser.

Karras T., et al. Elucidating the Design Space of Diffusion-Based Generative Models, 2022
Song J., Meng C., Ermon S. Denoising Diffusion Implicit Models, 2021

Recap of Previous Lecture

Now learn the velocity field itself. Recall $\mathbf{x}_t = \mathbf{x}(t)$ for $t \in [0, 1]$, with $\mathbf{x}_0 \sim p_0(\mathbf{x})$ (mostly $\mathcal{N}(0, \mathbf{I})$), $\mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$.

Time convention: unlike the diffusion sections, here $t = 0$ denotes noise and $t = 1$ denotes data. Sampling runs forward in time.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right)$$

$$\Leftrightarrow \quad \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

- Likelihood-based CNF training requires ODE solves and Jacobian trace evaluations.
- The adjoint method computes gradients through the ODE solution; repeated solves can be costly.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Recap of Previous Lecture

Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Flow matching fits velocities by regression, without model ODE solves or trace evaluations during training.

Outline

01 Conditional Flow Matching (CFM)

CFM Objective
Conditional Probability Paths

02 One-Sided Conditioning

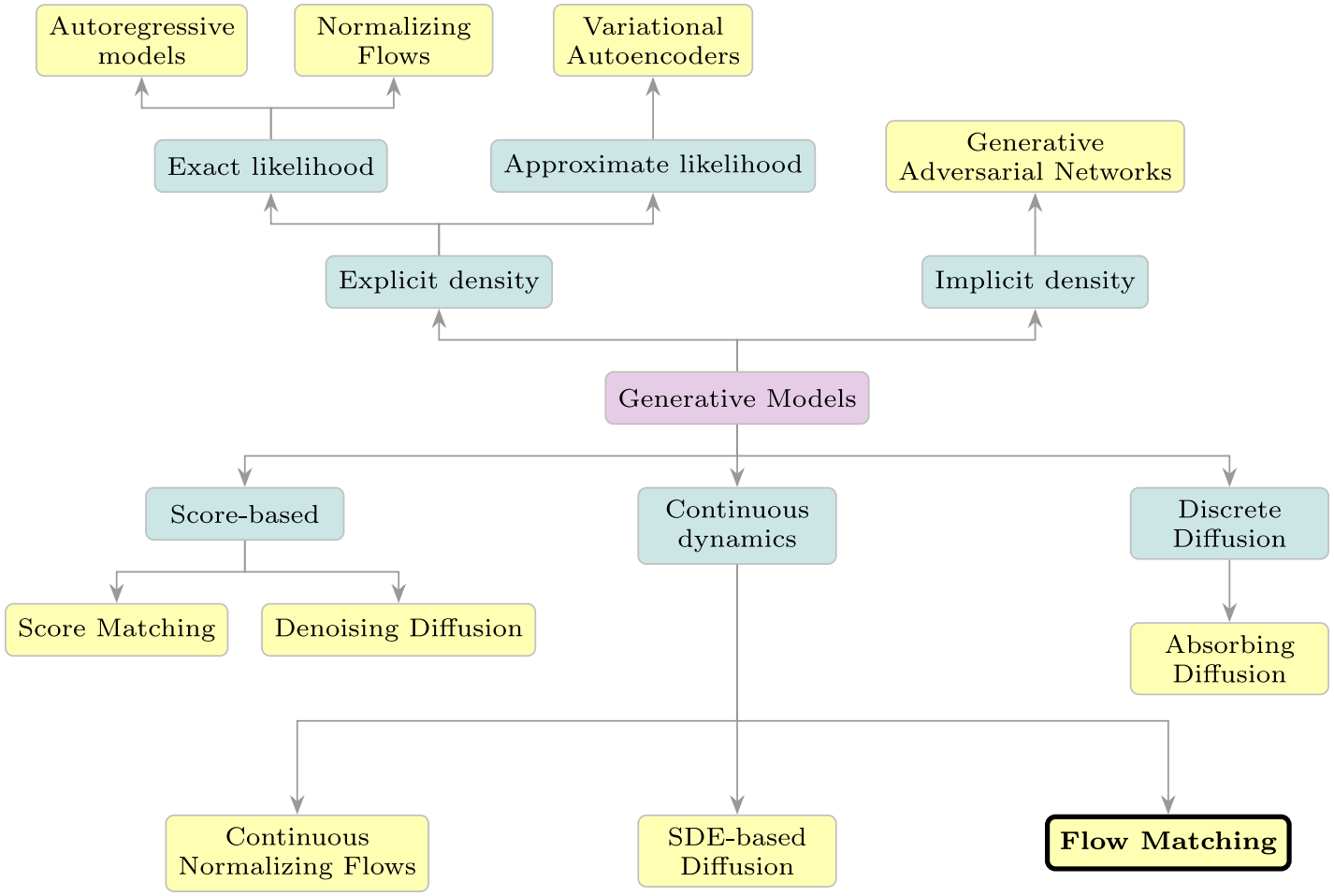
Outline

01 **Conditional Flow Matching (CFM)**

CFM Objective
Conditional Probability Paths

02 One-Sided Conditioning

Generative Models Taxonomy



Outline

01 **Conditional Flow Matching (CFM)**

CFM Objective
Conditional Probability Paths

02 One-Sided Conditioning

Flow Matching

Let's introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**.

The conditional probability path $p_t(\mathbf{x}|\mathbf{z})$ satisfies the KFP theorem:

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})\right),$$

where $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t) \quad \Rightarrow \quad \frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$$

Conditional Vector Field

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z})\right),$$

What's the relationship between $\mathbf{v}(\mathbf{x}, t)$ and $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$?

Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})} \mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$

Conditional Vector Field

Proof

$$
\begin{aligned}
\frac{\partial p_t(\mathbf{x})}{\partial t} &= \frac{\partial}{\partial t} \int p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \\
&= \int \left(\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} \right) p(\mathbf{z}) d\mathbf{z} \\
&= \int \left(-\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z}) \right) \right) p(\mathbf{z}) d\mathbf{z} \\
&= -\operatorname{div}\left(\int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) p_t(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \right) \\
&= -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x}) \right)
\end{aligned}
$$

Flow Matching

Flow Matching (FM)

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Theorem

If $p_t(\mathbf{x})>0$ for all $\mathbf{x}\in\mathbb{R}^m$ and $0\leq t<1$, the FM and CFM objectives differ by a constant independent of $\boldsymbol{\theta}$, so they have the same minimizers.

Proof

This can be proved in a similar way as in the denoising score matching theorem.

Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left\|\mathbf{v}(\mathbf{x},\mathbf{z},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

Proof

$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

$$= \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left[\left\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x},t)\mathbf{v}(\mathbf{x},t)\right] + \text{const}(\boldsymbol{\theta})$$

Conditional Flow Matching

Proof (continued)

$$
\begin{aligned}
&\mathbb{E}_{p_t(\mathbf{x})}\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] \\
&\quad= \int p_t(\mathbf{x})\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\left(\int p_t(\mathbf{z}|\mathbf{x})\mathbf{v}(\mathbf{x}, \mathbf{z}, t)d\mathbf{z}\right)\right]d\mathbf{x} \\
&\quad= \int\int p_t(\mathbf{x})p_t(\mathbf{z}|\mathbf{x})\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right]d\mathbf{z}d\mathbf{x} \\
&\quad= \mathbb{E}_{p(\mathbf{z})}\mathbb{E}_{p_t(\mathbf{x}|\mathbf{z})}\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right]
\end{aligned}
$$

Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p(\mathbf{z})}p_0(\mathbf{x}|\mathbf{z})=\mathcal{N}(0,\mathbf{I});\quad \mathbb{E}_{p(\mathbf{z})}p_1(\mathbf{x}|\mathbf{z})=p_{\text{data}}(\mathbf{x}).$$

Training

1. Sample $t\sim U[0,1]$, $\mathbf{z}\sim p(\mathbf{z})$, $\mathbf{x}_t\sim p_t(\mathbf{x}|\mathbf{z})$.
2. Compute loss $\mathcal{L}=\left\|\mathbf{v}(\mathbf{x}_t,\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\boldsymbol{\psi}_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

Marginal-to-Conditional: Scores vs Velocities

Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Optimal predictor:

$$\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right] = \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)$$

Conditional Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 \to \min_{\boldsymbol{\theta}}$$

Optimal predictor: $\mathbf{v}_{\boldsymbol{\theta}^*}(\mathbf{x},t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x},\mathbf{z},t) = \mathbf{v}(\mathbf{x},t)$.

In both cases, the optimal MSE predictor is the posterior mean of the **conditional** target, which equals the intractable **marginal** quantity (score or velocity).

Outline

01 **Conditional Flow Matching (CFM)**

CFM Objective
Conditional Probability Paths

02 One-Sided Conditioning

Conditional Flow Matching: Continuity Equations

Continuity Equations

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) \cdot p_t(\mathbf{x})\right)$$
$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) \cdot p_t(\mathbf{x}|\mathbf{z})\right)$$

What is the relationship between $p_t(\mathbf{x})$ and $\mathbf{v}(\mathbf{x}, t)$?

- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.
- But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t)p_t(\mathbf{x})) = 0$ can generate a valid probability path.
- Moreover, not every given path $p_t(\mathbf{x})$ can actually arise from particles moving under some velocity field $\mathbf{v}(\mathbf{x}, t)$.

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

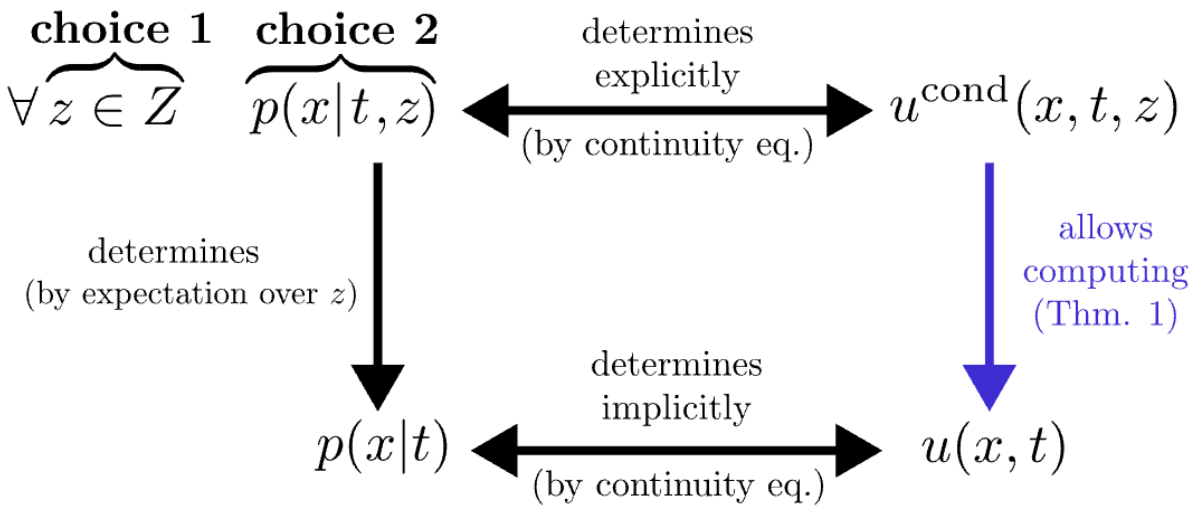
- $p_t(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$ uniquely determine $p_t(\mathbf{x})$.
- The reverse is not true: many different conditional families integrate to the same marginal.

Vector fields

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t)\frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})}d\mathbf{z}$$

- $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ and $p_t(\mathbf{x}|\mathbf{z})$ uniquely determine $\mathbf{v}(\mathbf{x}, t)$.
- But marginal velocity does not determine conditional velocities (many decompositions exist).

From Marginal to Conditional Paths



Open Questions

- **Q1** How should we choose the convenient conditioning latent variable $\mathbf{z}$?
- **Q2** How can we parametrize $p_t(\mathbf{x})$ (or $p_t(\mathbf{x}|\mathbf{z})$) to enforce the following constraints?

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

image credit: A Visual Dive into Conditional Flow Matching

Gaussian Conditional Probability Paths [Q2]

Let's consider the following parametrization:

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right)$$

The concrete $\boldsymbol{\mu}_t(\mathbf{z})$, $\boldsymbol{\sigma}_t(\mathbf{z})$ (which satisfy the boundary constraints) will be fixed once we choose $\mathbf{z}$ [Q1].

Gaussian flow

- Let's derive the vector field for **any** Gaussian path $p_t(\mathbf{x}|\mathbf{z})$.
- There are infinitely many flows $\boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z})$ (or, equivalently, vector fields $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$) that generate a particular probability path $p_t(\mathbf{x}|\mathbf{z})$.
- Let's consider the following flow:

$$\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon}$ is fixed by the condition: $\mathbf{x}_0 = \boldsymbol{\mu}_0(\mathbf{z}) + \boldsymbol{\sigma}_0(\mathbf{z}) \odot \boldsymbol{\epsilon}$.

Gaussian Conditional Vector Field

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right);$$
$$\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z}) = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

How to derive the expression for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$ for this particular flow $\boldsymbol{\psi}_t(\mathbf{x}_0, \mathbf{z})$?

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t); \quad \boldsymbol{\epsilon} = \frac{1}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))$$

$$\frac{d\mathbf{x}_t}{dt} = \boldsymbol{\mu}_t'(\mathbf{z}) + \boldsymbol{\sigma}_t'(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

$$\boxed{\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) = \boldsymbol{\psi}_t'\left(\boldsymbol{\psi}_t(\mathbf{x}_t, \mathbf{z})^{-1}, \mathbf{z}\right) = \boldsymbol{\mu}_t'(\mathbf{z}) + \frac{\boldsymbol{\sigma}_t'(\mathbf{z})}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))}$$

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02 One-Sided Conditioning

Endpoint Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Let's define our latent variable $\mathbf{z}$.

Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z}=\mathbf{x}_1$. Then $p(\mathbf{z})=p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x})=\int p_t(\mathbf{x}|\mathbf{x}_1)p_1(\mathbf{x}_1)d\mathbf{x}_1$$

We need to ensure the boundary constraints:

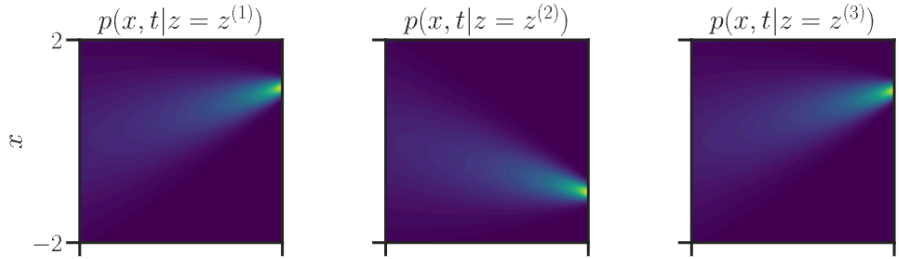
$$\begin{cases}\mathbb{E}_{p(\mathbf{z})}p_0(\mathbf{x}|\mathbf{z})=\mathcal{N}(0,\mathbf{I})\\ \mathbb{E}_{p(\mathbf{z})}p_1(\mathbf{x}|\mathbf{z})=p_{\text{data}}(\mathbf{x})\end{cases}\Rightarrow\quad\begin{cases}p_0(\mathbf{x}|\mathbf{x}_1)=\mathcal{N}(0,\mathbf{I});\\ p_1(\mathbf{x}|\mathbf{x}_1)=\delta(\mathbf{x}-\mathbf{x}_1)\end{cases}$$

One-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \quad p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_1)\right);$$
$$\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_1) \odot \boldsymbol{\epsilon}$$



Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases} \Rightarrow \begin{cases} p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2 \cdot \mathbf{I}\right) \\ \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \end{cases}$$

image credit: A Visual Dive into Conditional Flow Matching

One-Sided Conditioning: Conditional Vector Field

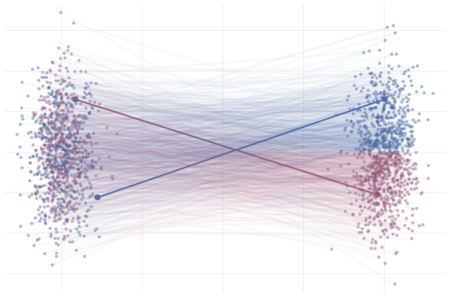
$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2\mathbf{I}\right); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \boldsymbol{\mu}_t'(\mathbf{x}_1) + \frac{\boldsymbol{\sigma}_t'(\mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_1))$$

$$\begin{aligned}\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) &= \mathbf{x}_1 - \frac{1}{1-t} \cdot (\mathbf{x}_t - t\mathbf{x}_1) = \frac{\mathbf{x}_1 - \mathbf{x}_t}{1-t}, \qquad t < 1 \\ &= \frac{\mathbf{x}_1 - t\mathbf{x}_1 - (1-t)\mathbf{x}_0}{1-t} \\ &= \mathbf{x}_1 - \mathbf{x}_0\end{aligned}$$

At $t = 1$, the quotient is undefined. Use the simplified training target $\mathbf{x}_1 - \mathbf{x}_0$, which contains no division.



The conditional vector field $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t)$ defines straight lines between $p_{\text{data}}(\mathbf{x})$ and $\mathcal{N}(0, \mathbf{I})$.

image credit: https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html

One-Sided Conditioning: CFM Objective

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

$$=\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{x}_1)}\left\|\frac{\mathbf{x}_1-\mathbf{x}}{1-t}-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2$$

$$=\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})}\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}\left(t\mathbf{x}_1+(1-t)\mathbf{x}_0,t\right)\right\|^2$$

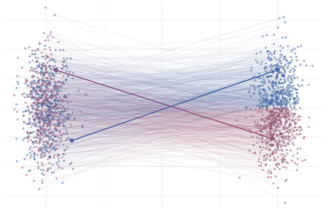
Rectified Flow

Regressing on $\mathbf{x}_1-\mathbf{x}_0$ along $\mathbf{x}_t=(1-t)\mathbf{x}_0+t\mathbf{x}_1$ gives the **rectified flow objective**.

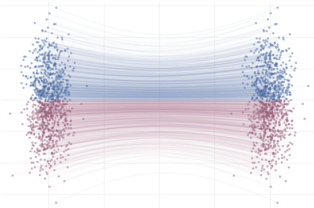
Straight training paths do not imply straight sampling trajectories. The optimal field averages conditional velocities:

$$\mathbf{v}_{\boldsymbol{\theta}^*}(\mathbf{x},t)=\mathbb{E}\left[\mathbf{x}_1-\mathbf{x}_0\mid\mathbf{x}_t=\mathbf{x}\right].$$

Conditional paths: straight



ODE trajectories: not necessarily straight



Lipman Y., et al. Flow Matching for Generative Modeling, 2022
Liu X., Gong C., Liu Q. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow, 2022

Rectified Flow: Training and Sampling

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})}\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

Training

1. Sample $\mathbf{x}_1\sim p_{\text{data}}(\mathbf{x})$, $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$, $t\sim U[0,1]$.
2. Compute noisy image $\mathbf{x}_t=t\mathbf{x}_1+(1-t)\mathbf{x}_0$.
3. Compute loss $\mathcal{L}=\left\|(\mathbf{x}_1-\mathbf{x}_0)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t,t)\right\|^2$.

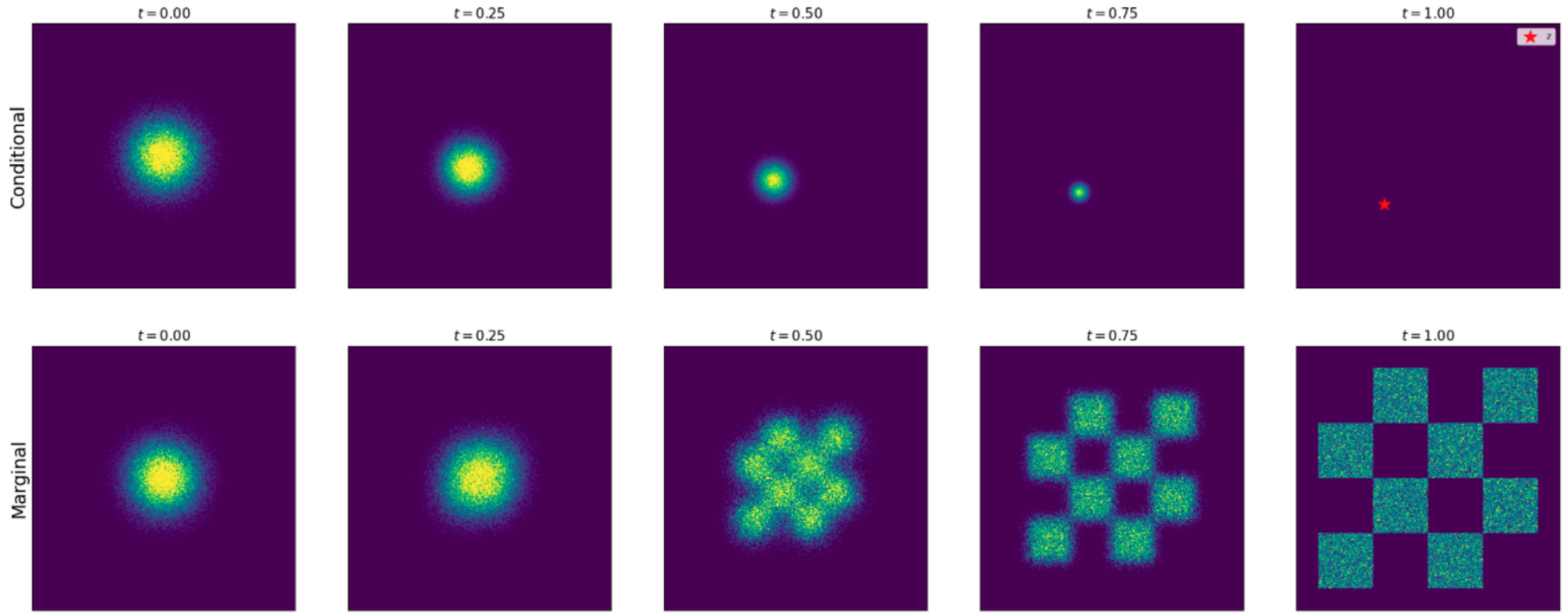
Sampling

1. Sample $\mathbf{x}_0\sim\mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1=\boldsymbol{\psi}_1(\mathbf{x}_0)=\texttt{ODESolve}_v(\mathbf{x}_0,\boldsymbol{\theta},t_0=0,t_1=1).$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022
Liu X., Gong C., Liu Q. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow, 2022

Conditional vs. Marginal Paths



- One-sided conditioning gives us the way to construct a generative model.
- Now we extend it to image-to-image formulation (mapping between two distinct data distributions $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$).

Summary

- Conditional flow matching introduces the latent variable $\mathbf{z}$, reformulating the initial task in terms of conditional dynamics.
- The marginal vector field can be expressed as the posterior mean of conditional vector fields: $\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})} \mathbf{v}(\mathbf{x}, \mathbf{z}, t)$.
- The FM and CFM objectives have the same optimal solution, making CFM a tractable alternative to FM.
- For Gaussian conditional paths $p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z}))$, the conditional vector field has a closed form for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$.
- One-sided conditioning $\mathbf{z} = \mathbf{x}_1$ with straight conditional paths yields the **rectified flow objective**, using the target $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \mathbf{x}_1 - \mathbf{x}_0$.